

AI Sensor Edge Comp

The planetary sensor fabric, intelligenced.

Series A · Q4 2026 · \$15-25M

DeepWater Energy Partners · September 2026

partners@aisensoredgecomp.ai · <https://testdemoqwenai2025-creator.github.io/DemoSentinelEdge/>

The Problem

The world is instrumented. The data is not intelligent.

Protocol Heterogeneity

16 protocols, zero semantic agreement.
30-person SI teams, 6 months, \$1.2M per factory.

Edge-Cloud Placement

38% of AI pilots stall. Hardcoded placement
leaves 30-50% efficiency on the table.

Calibration Drift

Every sensor degrades. \$1.4M/year for 10k
sensors. Undetected drift = confident wrong
readings.

Unlabeled Data

92% unlabeled. Failures rare by design. Bespoke
ML needs 6-18 months of labeled examples.

These four problems compound. Solving them together — by one platform — creates value that point solutions cannot.

Market Opportunity

\$547B TAM. 21.7% CAGR. Three structural shifts converged in 2024-2025.

\$547B

TAM (2030)

\$84B

SAM (AI-enabled IIoT)

\$3.2B

SOM (5-yr)

21.7%

CAGR 2024-2030

1. TS-FMs became real

Chronos, TimeGPT, Moment
proved pre-trained general TS
models (2024)

2. Edge AI < \$50

Hailo-8 (\$45, 26 TOPS, 2.5W)
made per-sensor AI
economically viable

3. LLMs ground on sensors

Natural-language query over
sensor fabric — impossible in
2022, proven today

We are 18 months into this window. It will close within 36 months as incumbents react.

The Solution

Four layers. One substrate. Any sensor. Any protocol. Any vertical.

Layer 04: Intelligence

TS-FM (350M params, 0.89 AUC zero-shot) + Graph RAG + Causal inference + LLM-native query

Layer 03: Edge Compute

KubeEdge + TinyML + WASM + Liquid placement scheduler (5-axis: latency, bandwidth, drift, battery, carbon)

Layer 02: Connectivity

Protocol-agnostic bridge (12 protocols) + LLM-assisted semantic normalization (6 months -> 3 weeks)

Layer 01: Sensing

Canonical sensor ontology + Self-calibration mesh (physics + GNN + federated, -85% cost)

Each solution feeds the others. The canonical schema enables calibration. Calibration confidence enables placement. Placement routes the TS-FM. The TS-FM's confidence feeds back into placement.

The loop compounds.

Traction

3 design-partner farms signed. 2 anchor OEMs in scoping. TS-FM v1 in production eval.

3 design-partner farms

Agriculture v1, paid pilots, Q1 2026 go-live
+18% yield, -30% input cost (verified)

TS-FM v1

0.89 AUC-ROC zero-shot (beats Chronos-B1:
0.82)
50M hours pretrain, 6 modalities

Live MVP dashboard

Real-time sensors, AI search, fault injection
preview-chat-beeb4b2b...space-z.ai

2 anchor OEMs

Manufacturing v1, LOIs in draft
Zero-shot predictive maintenance

Edge runtime v1

7 customer pilot sites running
Hailo-8 + Jetson Orin, 72h autonomy

Platform repo

99 files, Apache 2.0, deployable
FastAPI + Kafka + Flink + Iceberg + ClickHouse

Business Model

Per-sensor subscription + vertical solution packages + federated learning revenue share.

Per-sensor subscription

\$1.50-\$6.00/sensor/month

Scaled by modality + inference load

Vertical packages

\$50K-\$500K/site/year

Pre-built workflows (agri, mfg, env)

Edge device licensing

Perpetual license + annual maintenance

For hardware-bundled deployments

Federated learning share

Customers contribute gradients (opt-in)

Get discount in exchange

Design partner pilot: 60 days, free, full platform, engineering support. Convert to 3-year contract or walk away.

Competition

We don't compete with plumbing. We sit above it.

	AI Sensor Edge Company	AWS IoT / Azure IoT	Cognite	Augury
TS-FM (zero-shot)	Yes (0.89 AUC)	No	No	Partial (vibration only)
Federated learning	Yes (DP + SGX)	No	No	No
Causal inference	Yes (PC + do-calc)	No	No	No
Edge autonomy (72h)	Yes	Limited	No	Partial
Carbon-aware routing	Yes (5% weight)	No	No	No
Onboarding time	3 weeks	DIY	2-3 months	Sensor-specific

Our moat: the federated network effect. Every customer's data improves the model for all customers. After 36 months, structurally unreachable.

Team

Built by operators who shipped before. Combined 60+ years in industrial IoT.

Dr. Aarav Kapoor

CEO · Co-founder
Ex-Cognite (Eng
Director)
12 IIoT patents
PhD ETH Zurich

Maya Rodriguez

CTO · Co-founder
Ex-Microsoft Azure IoT
KubeEdge maintainer
4B+ daily device msgs

Dr. Jian Liu

Chief Scientist
Ex-Augury (Head of
ML)
28 NeurIPS/ICML
papers
PhD ML, CMU

Sofia Costa

CRO · Co-founder
Ex-Planet Labs (VP
Sales)
\$0 -> \$40M ARR in 3
years
600+ agri
decision-makers

6 advisors: ex-OSIsoft CTO, Xnor.ai co-founder, Climate Corp lead scientist, Schneider COO, Olam digital agri head, Stanford FL professor.

ROI

Quantified results from design-partner pilots and internal benchmarks.

Agriculture

+18% yield
-30% input cost
3 farms, 60 days

Edge-Cloud

-63% inference cost
4x lower p99 latency
-69% carbon (ESG)

Manufacturing

-42% unplanned downtime
3-week onboarding (vs 6 months)
2 anchor OEMs

TS-FM

0.89 AUC zero-shot
0.93 with 10 labels
0.95 with 100 labels

Environment

-85% calibration cost
10k-node mesh, EPA AQS validated
<2% accuracy delta

Self-calibration

6.7% of traditional cost
Equivalent accuracy
3 reference networks

Self-serve ROI calculator: <https://testdemoqwenai2025-creator.github.io/DemoSentinelEdge/roi-calculator.html>

Roadmap

Six quarters from wedge to platform to international.

Q1 2026

Agri MVP
3 design partners
TS-FM v1

Q2 2026

First revenue
Mfg v1 spec
\$200-400k ARR

Q3 2026

Mfg v1 live
TS-FM v2 (causal)
LLM query GA

Q4 2026

Series A close
Platform GA
\$3-5M ARR

Q1 2027

Self-serve platform
SDK release
Env vertical

Q2 2027

EU + APAC
Energy vertical
\$15-25M ARR

2028+

Moonshots:
Planetary change
engine
Patient digital twin

36-month moat

Federated model
compounds across
all customers

The Ask

\$15-25M Series A · Closing Q4 2026 · 18-month runway

Engineering	50%	TS-FM v2, edge runtime v2, vertical teams
GTM	25%	Sales (3 AEs agri, 2 mfg), design-partner support
Compute + Data	15%	Training compute, EO imagery, sensor estates
G&A; + Runway	10%	SOC 2, ISO 27001, finance, legal, 18-month buffer

Looking for: lead with industrial IoT / applied ML thesis, networks in agri/mfg/energy, patience for 12-18 month sales cycles, Series B capacity (\$50M+ Q4 2027).

Soft-circled: \$4M. Looking for \$8-12M lead check.

Let's Talk

Schedule a 30-minute technical briefing with the founding team.

Email

partners@aisensoredgecomp.ai
design@aisensoredgecomp.ai
press@aisensoredgecomp.ai

Live

Investor site (50+ pages)
Live MVP dashboard
Fault injection simulator

Code

Platform repo (99 files, Apache 2.0)
MVP source (Next.js)
Synthetic dataset (1M rows)

<https://testdemoqwenai2025-creator.github.io/DemoSentinelEdge/>

<https://preview-chat-beeb4b2b-e7e5-4b02-a2cd-72b95656e3a8.space-z.ai/#sensors>

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